

Siliclusterv3

Two-capacitor serial DAC

Simulation and validation companion | October 2, 2026 | Jonah Saunders, independent designer, United States

Prepared physical macro; organizer interface confirmation remains open. The 180 x 115 um SKY130 cell passes local physical and circuit checks. Three digital phase inputs for an analog slot, final signal pins, and power-grid landings still need organizer confirmation. No submission, acceptance, fabrication, or measured silicon is claimed.

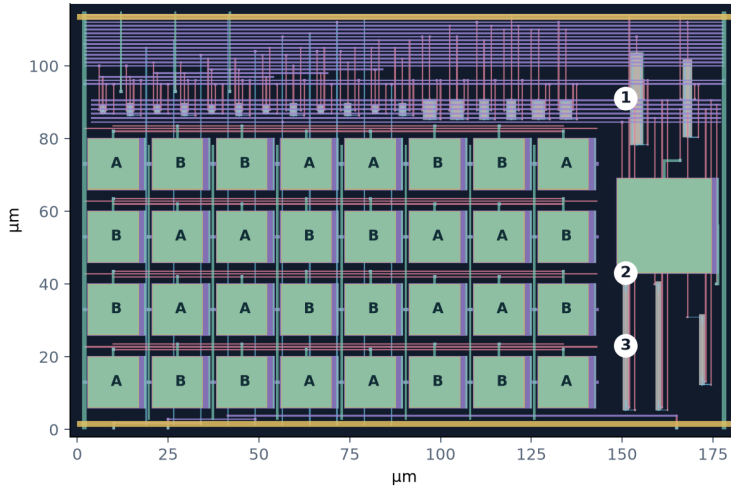
Final-layout check	Result
Size / area	180 x 115 um / 20,700 um ²
Area reduction from source	43.05%
Full DRC / antenna	0 violations / 0 feedback
Native and re-imported GDS LVS	Unique match; no property errors
Schematic ERC / visible connections	0 violations / 72 checked
Complete code sweeps	9 x 256; all strictly monotonic
Worst endpoint INL	0.5526 LSB
DNL range	-0.0313 to 0.5384 LSB
Stability cases / minimum margin	120 / 71.25 degrees
PVT probe points / mismatch seeds	30 / 32

Qualified assumptions: 1.8 V nominal supply; boundaries 1.62/1.98 V; five process corners; -40/27/85 C samples; fixed 0.2/0.9 V references; 5-20 pF load with 10 Mohm DC impedance. Assumed analog paths include 500 ohm series resistance and 5 pF on reference inputs. These are test fixtures, not confirmed organizer pad models.

Endpoint INL removes offset and gain. Accurate absolute voltage requires calibration. Discrete model-based tests do not establish silicon yield, ENOB, or final chip integration performance.

Circuit and compact floorplan

Silicluster v3 SKY130 GDS | 180 × 115 μm



1 TRANSISTORS AND ROUTING

M1-M6: phase inverters
M7-M12: reference and share switches
M13-M20: bias and output buffer

2 MATCHED CONVERSION BANKS

A = C1 / SAMPLE B = C2 / HOLD
16 × (14 × 14 μm) MIM units per bank
6.454 pF nominal per bank

3 BIAS AND COMPENSATION

R1-R3 and the separate C3 capacitor

Common-centroid placement

A	B	B	A	A	B	B	A
B	A	A	B	B	A	A	B
B	A	A	B	B	A	A	B
A	B	B	A	A	B	B	A

Both bank centroids: (72.75, 43.00) μm

Mask geometry from silicluster_jonah_2cap_dac.gds · SHA-256 befb7d48065a35ca...

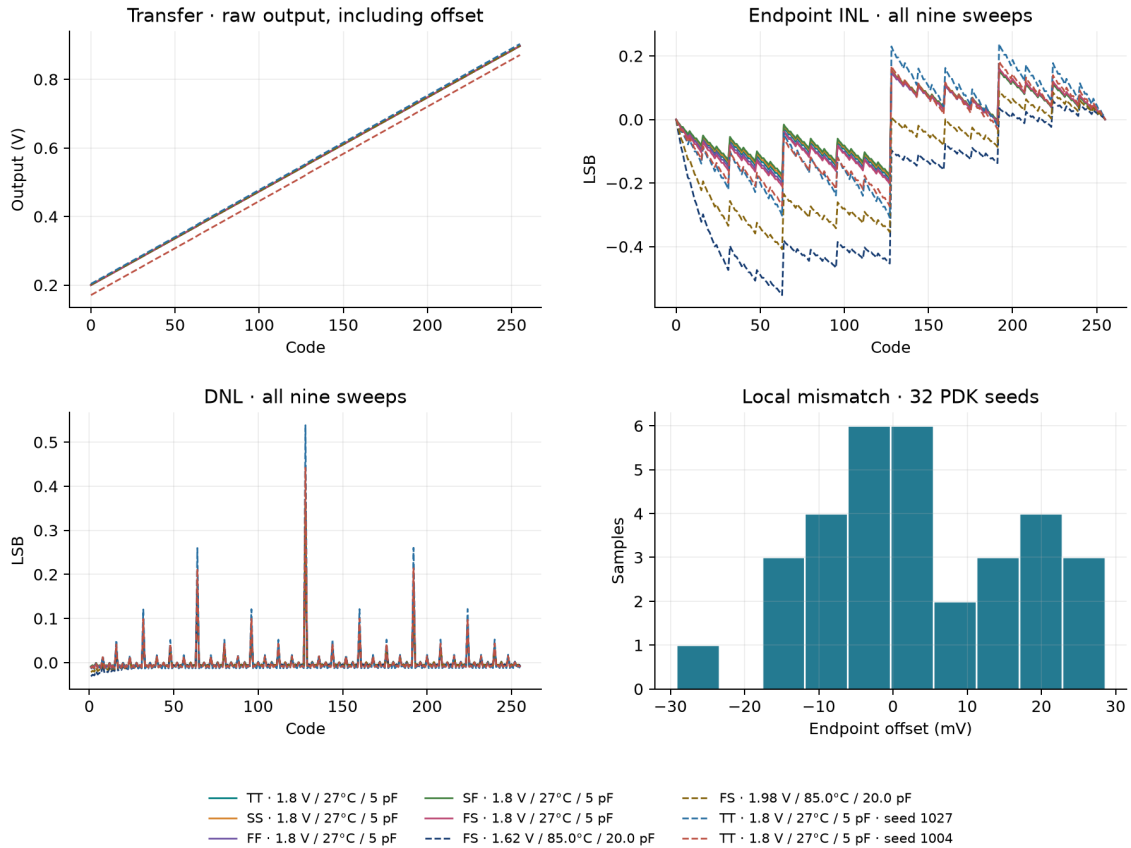
The conversion core shares charge between two nominally 6.454 pF banks, each sixteen 14 x 14 μm MIM units. Their common centroids are (72.75, 43.00) μm. Ground shields separate sample/hold routing. Twenty MOS transistors implement phase inversion, switching, bias, and a unity-gain output buffer. A separate 1.373 pF C3 capacitor compensates that buffer.

Device sizes and conversion capacitance are preserved from the source circuit. The native KiCad schematic and PDF are supplied separately for detailed reading. All MOS bulk ties are explicit in SPICE and silicon. Vertical M4 power rails and horizontal M5 rails implement the call's grid orientation; all geometry is inside the declared boundary on a 5 nm grid.

Interface: VGND, VDPWR, charge_high, charge_low, share, vref_high, vref_low, vout. Signal pins are provisional M4 rectangles. The call grants two analog inputs and two analog outputs, but does not explicitly allocate digital controls to analog projects.

Transfer, linearity, and mismatch

Post-layout results from the final RC extraction
2,304 code measurements · nine monotonic sweeps · worst endpoint INL 0.55 LSB



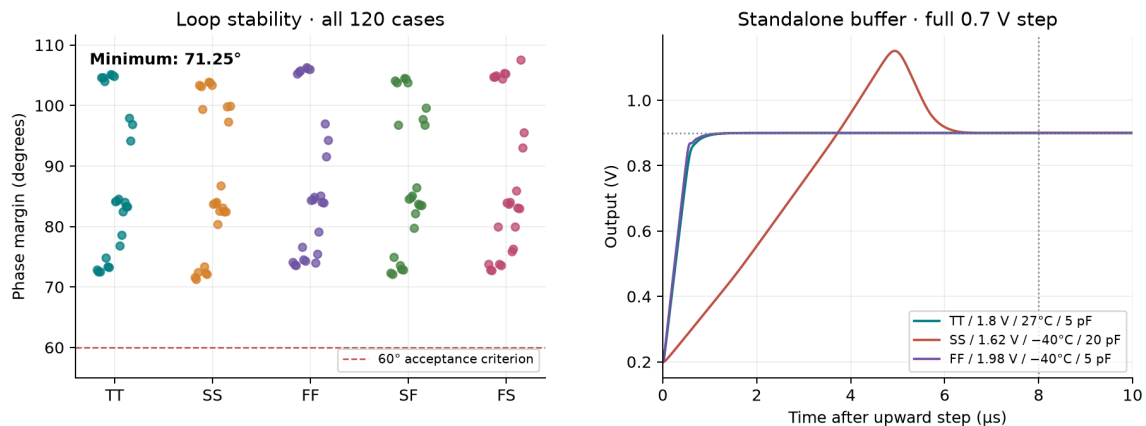
Five full sweeps cover nominal supply and temperature at TT, SS, FF, SF, and FS. Two additional full sweeps use PVT probe conditions with the highest observed selected-code INL/DNL; two use mismatch seeds with the highest observed selected-code INL/raw error. Selection comes from this layout's results and is recorded in `full_configs.json`. These are sampled stress cases, not an exhaustive worst-case search.

Across all nine sweeps, worst endpoint INL is 0.5526 LSB. Thirty PVT points and 32 local-mismatch seeds also pass selected-code monotonicity. Mismatch endpoint offset spans -29.22 to 28.59 mV. Calibrate codes 0 and 255 to remove measured endpoint offset/gain; residual nonlinear error remains.

Each measured code begins with the same initialization. Eight codes are grouped per run. All 2,304 complete-sweep measurements are supplied in `all_code_results.csv` with detailed case JSON and simulator logs. Local mismatch sampling excludes wafer-scale systematic capacitor gradients and package variation.

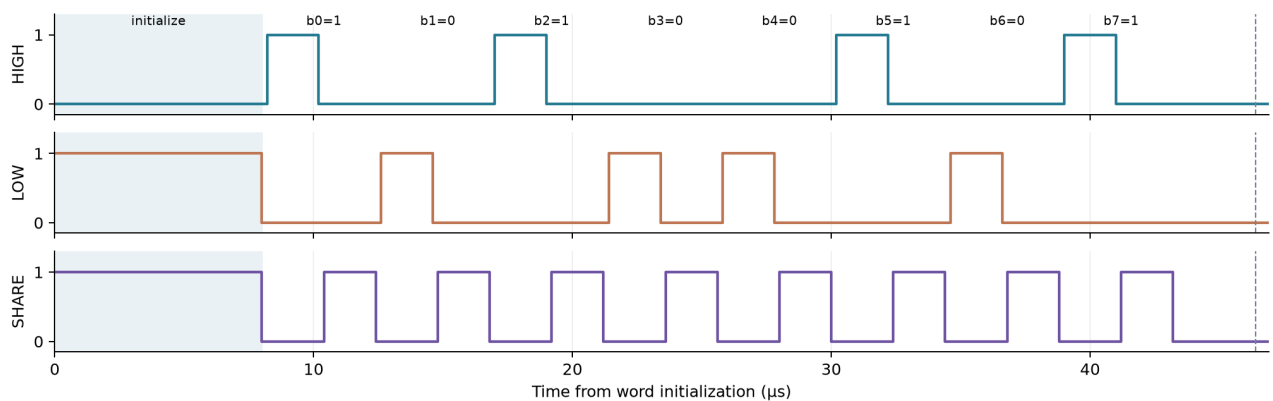
Timing, stability, and retention

Stability and large-signal settling are separate checks



Worst standalone upward step: 6.41 μs to $\pm\frac{1}{2}$ LSB; allow 8 μs. The DAC protocol is verified separately.

Phase timing example · code 0xA5, LSB first



Protocol illustration, not a simulated waveform. Read at 46.4 μs; 0.5 μs gap gives 46.9 μs/word.

All 120 buffer feedback-injection cases exceed 60 degrees phase margin; the minimum is 71.25 degrees. Worst full-range standalone buffer settling is 6.41 μs to half an LSB; allow 8 μs. Worst overshoot is 250.9 mV. The actual DAC read timing is tested separately by complete conversion sweeps.

Hot hold cases show up to 0.2896 LSB of immediate feedthrough when the sample bank changes to the opposite reference, followed by at most 0.00305 LSB drift over 50 μs. Refresh periodically. Integrated modeled buffer noise is 57.4-58.3 μV RMS over 0.1 Hz-10.66 kHz; kT/C at 85 C is estimated at 27.7 μV RMS. These are not measured ENOB results.

Evidence, reproduction, and submission

Physical evidence includes native layout DRC, full DRC after independent GDS re-import, antenna checks, and native/re-imported transistor-level LVS. The geometry audit checks bounds, manufacturing grid, capacitor centroids, and all eight GDS/LEF pin rectangles. A fresh build reproduces the circuit and all GDS polygons. Preserve resistor URPM mask hints when importing GDS.

Coupled-C and full RC extraction use the pinned IIC OSIC Tools 2026.07 image with SKY130A. Resistance threshold/minimum is 1 ohm and delay cutoff is zero. Exact reduction removes only capacitance-free resistor nodes. Raw/reduced RC, cached/full models, and 100 ns/25 ns maximum time steps agree within 10 uV at four representative codes. Repeating a mismatch seed gives identical output. Source/model/result hashes are audited before reporting.

Reproduce from the repository root with the pinned image and bash scripts/verify.sh. The --smoke variant runs physical checks, a fresh build, and a ten-code extracted transfer probe. Full qualification regenerates all sweeps and reports. See docs/reproduce.md for Docker and native KiCad commands. The complete narrative validation document is verification/verification.md.

Package: final GDS, Magic top and child cells, LEF, external-interface Verilog, editable schematic, reference and extracted netlists, testbenches, PDK model cache and license, simulator results/logs, DRC/LVS evidence, figures, generators, and a hash manifest. Redundant raw waveform vectors and DUT snapshots are regenerated by scripts.

Remaining integration work: obtain organizer approval for three digital phase inputs on an analog slot and the physical pin/power-rail template. Incorporate any required changes and rerun checks. Fill the participant city before sending the prepared submission email with the current ZIP. Country is United States and affiliation is independent designer. No email has been sent.

The supplied Siliclusterv3 full call, sections 8-11 and 16, defines the analog I/O limits, size, required ZIP contents, grid orientation, and participant information. It gives a deadline of October 31, 2026, 11:59 p.m. Mexico City time, subject to remaining slots and compliance. Acceptance does not guarantee a packaged chip. The call PDF is included under requirements/.

Repository: https://github.com/jonahsaunders/Siliclusterv3_2_cap_DAC

GDS SHA-256: befb7d48065a35ca57e100f0ab284685
9c1f49a87afc503393072093e10a78d2

Reduced RC SHA-256: cc6a3b5c6d07e2b634c93991ac60591b
d67764222b210e165198f63464744e6c